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Emerging technologies for efficient water use in agriculture: A review of current trends and future directions

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ABSTRACT

Water sets our planet apart from all others we know of. Although the global supply of fresh water is sufficient to meet current and future water demands, its distribution in terms of location and timing is uneven. Many regions face shortages of freshwater resources and cannot fulfill domestic, economic, and environmental needs. In these areas, the scarcity of clean water for drinking and sanitation hampers human health and productivity, thereby limiting economic growth and the preservation of a clean environment and healthy ecosystems. Currently, 7 % of the global population resides in water-scarce regions, and this situation is expected to worsen as the world's population reaches 9 billion by 2050, with 67 % living in urban areas. Efficient water use in agriculture is vital for food security, environmental protection, and economic viability. Conventional water management methods are inadequate for addressing emerging challenges, necessitating advanced technologies such as sensors, artificial intelligence, remote sensing, nanotechnology, and desalination. Innovative technologies offer real-time monitoring and precise resource distribution to improve water management, enhancing efficiency and sustainability. However, widespread adoption faces challenges such as cost, regulatory issues, technological complexity, and infrastructure limitations. Addressing data privacy, cybersecurity, and equitable access to technology is also crucial. This article provides comprehensive coverage of these innovative technologies and tools, considering the operational characteristics of emerging technologies and the existing technology gap. Overall, this study examines innovations in water management technology, highlights ongoing research, identifies gaps, and offers recommendations for policymakers, water utilities, and researchers.

1. Introduction

Water scarcity is a critical issue that arises when there is an insufficient supply of fresh water to meet the demand [1,2]. The agricultural industry is the biggest water consumer in the world, utilizing a staggering 80–90 % of all water consumed. With an average water use efficiency of only 45 % and a concerning water loss of over 50 %, this sector could use more water [3]. Around 7 % of the global population resides in areas with insufficient water supplies. Severe scarcity of water is faced by around 4 billion people [4]. By 2050, around 67 % of the world's population will live in urban areas [5]. With a growing world population, climate change constantly challenges food supply and

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demand. The world's population is predicted to reach 9 billion by 2050, nearly a 25 % increase over the present number [6]. Feeding a growing population in the face of depleting and overusing water resources is one of the main issues we face today [7]. Using water efficiently for sustainable agriculture production is imperative [8]. It is vital for food security, conserves water resources, protects the environment, and improves economic viability [9]. Prioritizing water efficiency in agricultural techniques can establish a more resilient and sustainable food system that can meet the needs of both current and future generations [10]. However, conventional water management techniques are insufficient for emerging challenges, hence the necessity of adopting advanced technologies. The concept of emergent technologies in water management includes a variety of novel ideas that intend to make water resources more available, sustainable, and efficient. Advanced sensors, artificial intelligence, remote sensing, nanotechnology, and desalination processing methods are among these technologies [11,12]. The application of such technologies makes for the real-time monitoring and exact distribution of resources with the help of innovative techniques and water management solutions.

In short, applying new technology is the key to overcoming the intricate problems of water management through improved monitoring and data gathering, better resource use, pollution control, climate change adaptation, community participation, and transparency [13–15]. To ensure the availability of clean and dependable water in the future, water managers could implement innovative technologies that would improve the efficiency, effectiveness, and sustainability of water-related solutions. With a selection of promising applications for many emerging technologies, several challenges exist to the widespread adoption of these technologies, for example, cost constraints, regulatory barriers, technological complexity, and inadequate infrastructure [16–20]. Moreover, the issues of data privacy, cybersecurity, and unshackled access to technology should be addressed to ensure inclusive and sustainable water management practices [21]. This review will focus on innovations in water management technology and the problems mentioned above as examples to show how these new technologies can be applied. Considering the operational characteristics of emerging technologies and a technology gap, this article provides a well-rounded coverage of innovative technologies and tools [22]. The area of consideration covers (but is not limited to) intelligent water systems, remote sensing, advanced water treatment methods, and sustainable water infrastructure [23,24]. This review paper will be accomplished by summarizing the ongoing research and stating the gaps. The aim is to identify research directions and provide recommendations for policy-makers, water utilities, and researchers in water management.

2. Smart irrigation systems

Different smart and innovative irrigation systems are used in water management in agricultural practice to reduce the wastage of water and save water in a sustainable way [25,26]. Irrigation controls with sensors precisely track the temperature and moisture content of the soil, maximizing water flow and minimizing waste [27]. By combining soil sensors and meteorological data, automated irrigation systems generate dynamic schedules that save water and increase crop productivity [28]. Drip and micro-irrigation techniques minimize losses from evaporation and runoff by directly delivering water to plant roots [29]. Water conservation and agricultural yield are enhanced by variable rate irrigation (VRI) systems, which electronically modify the amount of water applied to different field areas [30]. These innovative approaches offer sustainable and cost-effective solutions for modern agriculture.

2.1. Sensor-based irrigation controllers for precise water application

Smart irrigation systems measure and estimate the reduction in plant moisture [31]. This allows the system to resupply water as needed while avoiding overwatering. Tensiometric and volumetric approaches, which are relatively simple, are used in soil moisture-based irrigation control. These values are coupled by a soil type-specific soil water characteristic curve. Regular maintenance is also necessary for the sensors to function properly. A moisture sensor is used by an intelligent automatic plant irrigation system to concentrate on regularly watering plants without human monitoring [32]. The operational model continuously monitors the soil's temperature and moisture level. Temperature and soil moisture can be regulated within a specific range based on crop type and soil composition. When the soil's moisture content or temperature departs from the predetermined range, the irrigation system is either turned on or off. The irrigation system will turn on and irrigate the plants with water when the soil temperature is high and the soil is dry. Fig. 1 shows the diagram of the smart irrigation system. The ATmega328 microprocessor, which powers the system, is its central component. The input pins of the controller are connected to the temperature and moisture sensors [33]. The servo motor and water pump are connected to the output pins. When the sensors deviate from the predefined range, the controller activates the pump. The servo motor's control over the pipe's angular position allows water to be spread evenly across the soil. The LED light indicates the state of the pump.

It may be further shown that this technique is much more beneficial when it is used extensively for agricultural reasons. Because of the current circumstances and water constraints, it is essential to select the most suitable irrigation schedules, especially in farms [34,35].

2.2. Automatic scheduling of irrigation utilizing soil moisture sensors and weather data

Automated irrigation scheduling using the data of weather and soil moisture sensors is a smart and efficient way to optimize water usage in agriculture and landscaping. Different smart irrigation systems used in the application of agriculture are included in Table 1. The discussion below and Fig. 2 show the typical working procedure of automated irrigation scheduling.

2.2.1. Weather data collection

Information about the weather, including temperature, humidity, wind direction, and precipitation, is gathered from online or local weather stations [36].

2.2.2. Soil moisture sensors

In the irrigated region, sensors are installed in the ground at different points that can determine soil moisture [37]. By measuring the soil's moisture content, these sensors can determine whether the soil is wet or dry.

2.2.3. Irrigation controller

An irrigation controller is the brain of the system. It calculates when and how much to irrigate based on data it gets from weather forecasting stations and the sensors on the ground.

2.2.4. Data processing and analysis

The irrigation controller uses data from soil moisture sensors and weather stations to determine the water requirements of plants [38]. Variables such as crop type, soil type, growth stage, and evapotranspiration rate are analyzed.

2.2.5. Irrigation schedule generation

The irrigation controller generates a watering schedule based on meteorological data and levels of soil moisture that can be easily adjusted to accommodate changes in plant requirements or weather patterns.

2.2.6. Automated irrigation

The irrigation controller turns on automatically based on a preset schedule. This process includes using drip emitters, sprinklers, or other

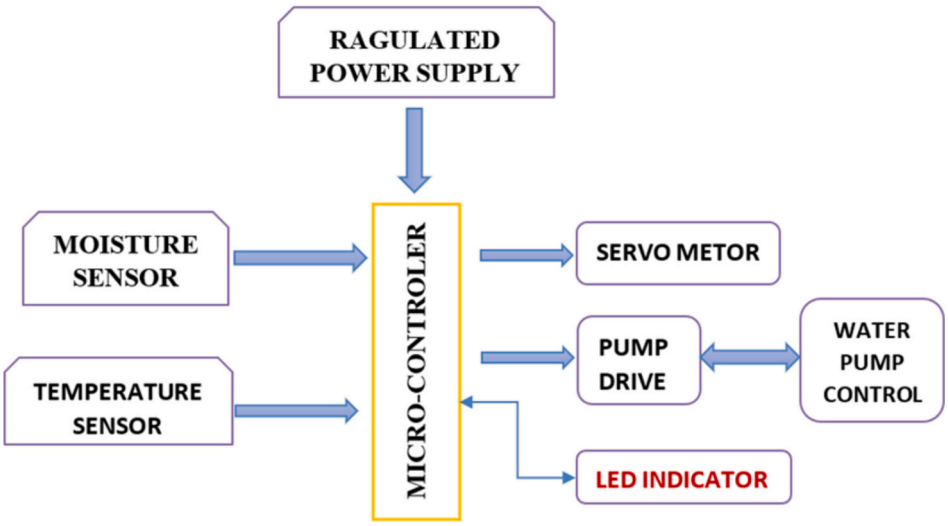


Fig. 1. A smart irrigation system block diagram.

Table 1
A table on different smart irrigation system used in Agriculture.

Irrigation system	Examples	Application	Merits	Drawbacks	References
Sensor-based irrigation	Soil Moisture Sensors (Vegetronix VH400 Soil Moisture Sensor)	Apply to avoid overwatering and limited irrigation according to the crop	Most suitable irrigation schedules and can be regulated within a specific range without help of human-being	Necessary of enough technical knowledge and technological cost	[32,38]
Weather-based automatic scheduling of irrigation	Rachio Smart Sprinkler Controller, Hunter Rain-Clik Sensor	To detect the data of weather and soil for smart irrigation	Water conservation, application of proper quantity of water, increasing yield, reduce labor cost	Proper knowledge and skill of maintaining the tools	[39,40]
Drip and micro-irrigation	Netafim Drip Emitters, Rain Bird Drip Line	To reduce water-loss at the time of irrigation	More efficient, direct water delivery to the root-zone	Management and setup cost, minimum technical knowledge needed	[29,44]
Variable rate irrigation (VRI)	Self-propelled rain gun sprinklers	To apply water according to the application and precision map	Economically beneficial and reduce the water loss, leaching loss	Require expertise to install, operate, and maintain	[52,54]
Smart Wi-Fi controllers	Orbit B-hyve Smart Wi-Fi Controller	To control and regulate the irrigation system from a certain distance	Less labor cost, reduce continuous physical appearance in to the crop field	Installation and operating knowledge and ability	[40,160]
Sprinkler systems with adjustable nozzles	Toro Precision Series Spray Nozzles	Controlled nozzles help to equal distribution of water into the field	Required irrigation water supply, less irrigation water waste	Mechanical and technical problem and cost	[135,224]
Evapotranspiration (ET) controllers	Rain Bird ESP-TM2 Controller	Use of soil-moisture sensor (SMS) systems to provided better irrigation efficiency and adequacy	Automated temperature control using sensors and microcontrollers, enhancing environmental conditions	The controller may not offer the flexibility to easily adapt to changing weather conditions or specific plant requirements	[7,136]
IoT-based irrigation systems	Netro Sprite Smart Sprinkler	It makes sure plants get the right amount of water by using sensors and microcontrollers to monitor ambient factors and soil moisture levels	Water conservation, increased crop yield, cost-effectiveness, remote monitoring and control	Initial setup cost, internet connection, technical problem	[35,185]
Flow meters	Flume 2 Smart Water Monitor	Providing an affordable, remotely monitored option	Highly accurate for calculating water use and flow, real-time monitoring	Maintenance cost and costly and vulnerable	[22]

watering methods.

2.2.7. *Monitoring and feedback*

The irrigation system ensures that water is used efficiently by monitoring weather conditions and soil moisture levels. Users can receive reports or notifications to track their water consumption and make adjustments accordingly.

The automated irrigation scheduling using weather data and soil moisture sensors is effective and advantageous. This irrigation system contributes to water conservation by ensuring that water is only used when required and in the proper quantity. When plants receive the proper quantity of water at the appropriate time, their growth is healthier and their yields are increased [39]. Automated systems save time and labor costs by reducing the need for irrigation schedule monitoring and modification [40]. The technology increases the overall efficacy and efficiency of water consumption by utilizing real-time data to guide irrigation decisions [41]. Overall, automated irrigation scheduling that utilizes weather data and soil moisture sensors is an indispensable tool for modern agriculture and landscaping. It helps to maximize productivity while minimizing water waste, making it an environmentally friendly and cost-effective solution [42,43]

2.3. *Drip and micro-irrigation techniques for minimizing water losses*

The use of drip and micro-irrigation systems is essential for reducing agricultural activities' water loss. By delivering water straight to the plant's root zone, these techniques reduce common losses such as runoff, deep percolation, and soil erosion [44]. These systems minimize

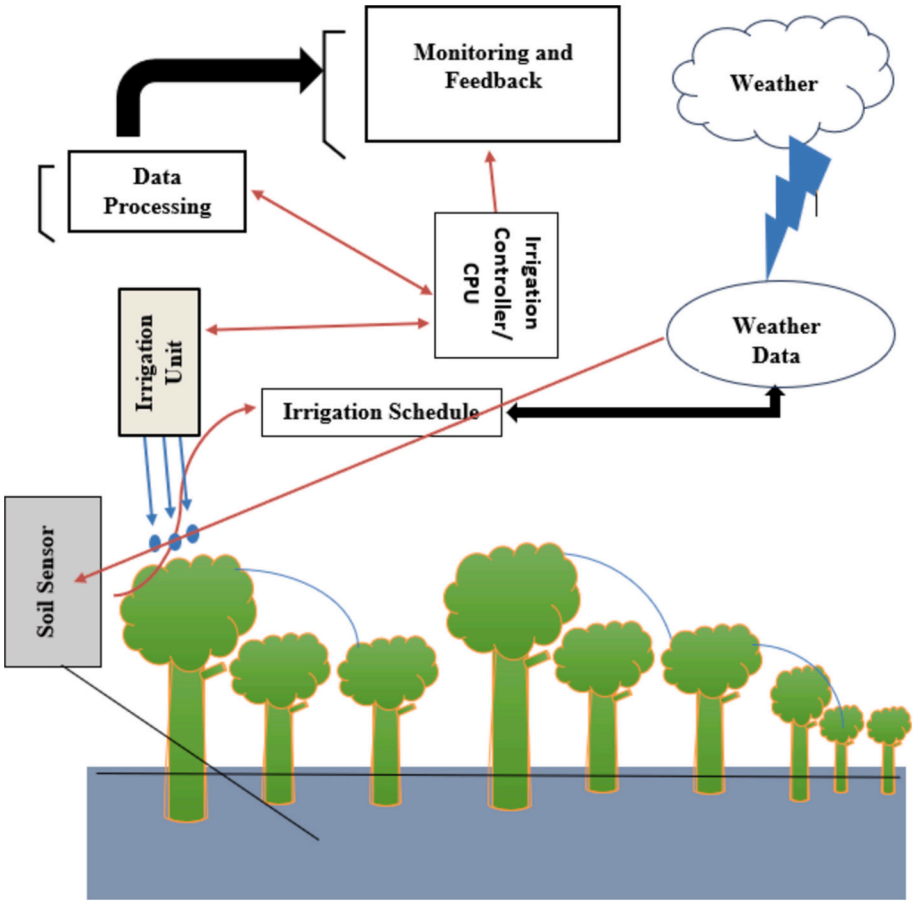


Fig. 2. Automated irrigation scheduling.

traditional losses such as soil evaporation by limiting watering to the consumptive use of plants and delivering a tiny, uniform flow of water at a constant discharge [45]. Micro-irrigation systems are more efficient than traditional surface irrigation for conserving and maximizing water usage. Micro-irrigation systems can be classified into two main types, such as drip irrigation systems and sprinkler irrigation systems. Drip irrigation applies water directly to plant roots using drippers or emitters at frequent intervals. Water is delivered drop-by-drop or through a microjet at a rate lower than the soil's infiltration rate. This method promotes efficient use of water and healthy plant growth. Types of drip irrigation systems are surface drip irrigation systems, subsurface drip irrigation, continuous flow drip irrigation, family drip, online drip, in-line drip, etc. [46]. A part of a drip irrigation pump; filtration systems, such as a water filter or sand filter (separator); fertilization systems (fertilizer mixed with irrigation water is called “fertigation”); pressure controller (pressure control valve/regulator); backwater flow prevention unit; large pipe and pipe fittings (mainline pipes); hydraulic control valves and safety valves; smaller diameter poly-tube, known as “laterals”; poly fittings and accessories to make drip connections; emitters or drippers; micro spray head; in-line dripper or in-line drip tube [47,48]. Sprinkler irrigation applies water to crops like natural rainfall. It uses pipes, pumps, and sprinklers to deliver water, with an operating pressure of 2.5 bar and a water supply rate exceeding 1000 lph. The sprinklers break water into small droplets, covering an area larger than 10 m. Types of sprinkler irrigation systems are center pivot, towable pivot, rain gun, impact sprinkler, pop-up sprinkler, linear move sprinkler, etc. [49]. Water sources of this irrigation system are canals, tube wells, open wells, etc. Centrifugal, submersible pumping units, and components of this system are sprinklers, riser pipes, main and lateral pipe lines, elbows, gauges, valves, and filters, among some other accessories [50,51].

2.4. Implementing variable rate irrigation (VRI) systems

Variable rate irrigation (VRI) utilizes electronic controls to adjust the speed of the center pivot and cycle sprinklers. This allows for the application of water to be adjusted along the entire length of the pivot. Depending on the level of precision desired by the farmer, sprinklers are typically regulated in groups of two to ten. Most existing pivots can have VRI installed retroactively. However, the cost of installation depends on the brand, the length of the pivot, and the level of resolution required by the farmer to address the field variability. These costs can vary greatly across different brands. The application rates are determined based on an application or prescription map. The merits of the VRI system are economic benefits by reducing wastage of water, pumping cost, leaching loss, surface runoff, and improvement of water conservation, crop water productivity, increasing revenue, and improving production techniques. On the other hand, the demerits are increased cost, payback period, labor input, risk of yield reduction and system breakdown, etc. [52–55].

3. Water-efficient crop varieties and breeding techniques

3.1. Development of drought-tolerant and water-efficient crop cultivars

The development of drought-tolerant and water-efficient crop cultivars is crucial for ensuring food security in the face of climate change-induced challenges. Breeding for drought-resistant crop varieties has been identified as a cost-effective and sustainable strategy to adapt crop production systems to increasingly harsh environmental conditions [56,57]. Selecting cultivars with better drought or heat stress tolerance, or breeding for stress tolerance, has been highlighted as a key approach to enhancing crop productivity under adverse conditions [58,59]. This

strategy is essential for sustaining the income of farmers and ensuring global food security [60]. Characterizing plant physiology has underscored the significance of developing drought-tolerant cultivars as the most effective method to address current challenges, offering farmers a relatively inexpensive and manageable crop production approach [61]. Breeding for drought-tolerant cultivars has been recognized as an economic and sustainable mitigation strategy against current and projected drought stress in crops like sorghum [62]. Furthermore, the development of drought-tolerant and climate-resilient cultivars has been emphasized as essential for sustaining farmers' income and global food security [60]. Efforts to develop drought-tolerant crop genotypes are essential components of drought mitigation strategies, aiming to enhance seed yield under drought stress conditions [63]. The use of biotechnological tools, such as marker-assisted selection and omics technologies, holds promise for advancing breeding efforts to produce climate-resilient crop varieties [64]. Additionally, the identification of drought-related quantitative trait loci (QTLs) and the application of transgenic technologies offer avenues for improving drought tolerance in crops [65]. In the context of drought stress, selecting cultivars that can maintain economically significant yields under water-deficient conditions is a top priority for crop production [66]. Root phenotyping has been identified as a valuable method for selecting drought-tolerant cultivars, particularly in crops like soybean, under field conditions [67]. The selection of more drought-tolerant cultivars with higher yields under stress conditions can be facilitated by specific characteristics such as wilting index and root morphological traits [63]. Breeding for drought tolerance is a complex process, especially considering the simultaneous challenges posed by various abiotic stresses like high temperatures, high irradiance, and nutrient imbalances [68]. Water use efficiency (WUE) is a critical parameter influencing crop yields, with greater WUE often associated with enhanced drought tolerance in crops like soybeans [69]. Sustainable forage and grass production during droughts and heat waves necessitates the development of drought-tolerant genotypes to ensure agricultural resilience [70]. Breeding for stress tolerance, utilizing biotechnological tools, and selecting cultivars with specific traits are key strategies to enhance crop productivity under adverse environmental conditions. By focusing on the development of drought-tolerant varieties, agricultural systems can become more resilient and sustainable in the long term.

3.2. Genetic engineering approaches for enhancing water use efficiency

Genetic engineering offers significant potential for enhancing water use efficiency in crops, a crucial factor in sustainable agriculture. By manipulating key regulators of stomatal movement, such as those involved in signaling pathways like ABA, ROS, and Ca^{2+}, it is feasible to increase water use efficiency in various crops [71]. This approach is supported by the high conservation of these signaling pathways across different plant species. Additionally, the selection for increased plant water use efficiency (WUE plant) is recognized as a crucial trait in breeding programs for food and biofuel crop species [72]. Through genetic engineering, it is possible to introduce novel traits into plants, such as improved water utilization and fertilizer efficiency, metabolic engineering for enhanced nutritional value, and adaptations for increased biomass production for biofuel purposes [73,74]. Studies have identified genetic components that influence root characteristics, offering potential avenues to enhance water use efficiency and mitigate crop losses under adverse environmental conditions [75]. Manipulating stomatal density through genetic interventions presents another avenue to enhance plant water use efficiency, thereby promoting bio-water conservation [76]. Efforts to genetically engineer key enzymes of the C_4 pathway in C_3 plants aim to boost photosynthetic efficiency, water use efficiency, and nitrogen utilization, although challenges remain in achieving desired metabolic flux alterations [77]. Furthermore, a deep understanding of transport mechanisms in guard cells provides a foundation for reverse engineering stomatal responses to enhance water use efficiency and crop yields [78]. The application of genetic engineering in crops extends beyond water use efficiency to address various challenges in agriculture. Introducing crassulacean acid metabolism (CAM) into C_3 crops is proposed as a strategy to increase water-use efficiency in plants [79]. Modulating abscisic acid receptors and coreceptors offers a means to enhance plant water use efficiency and productivity without compromising growth and yield [80]. Genetic manipulation presents opportunities to improve agronomic performance, pest resistance, and end-use quality of crops by incorporating specific genes [81]. While genetic engineering is a powerful tool for enhancing crop traits, it complements rather than replaces conventional breeding methods, contributing to the overall efficiency of crop improvement [82]. Strategies for enhancing salt tolerance through genetic engineering are being explored as a viable approach for crop enhancement [83]. Advances in understanding plant physiology and agronomy have led to improvements in water productivity in crops, emphasizing the importance of continued research in this area [84]. Efforts to enhance crop productivity under water-limited conditions underscore the significance of intrinsic genetic improvements as a sustainable solution [85]. Manipulating stomatal resistance to reduce transpiration losses represents a potential avenue for improving water use efficiency in crops under limited water conditions [86]. The optimization of genetic transformation methods in crop plants is crucial for developing desired traits to meet future agricultural demands and changing environmental conditions [87]. Genetically engineered crops, when integrated with optimal management practices, have been shown to enhance food production sustainability [88]. Modeling crop responses to water inputs is essential for regulating field water regimes and improving water use efficiency in agriculture [89]. So, genetic engineering offers a powerful approach to enhance water use efficiency in crops, addressing the challenges of water scarcity and sustainable agriculture. By targeting key regulators of stomatal movement, introducing novel traits, and optimizing plant responses to water, genetic interventions can play a pivotal role in ensuring food security and environmental sustainability.

3.3. Marker-assisted selection (MAS) for breeding drought-resistant crop varieties

Marker-assisted selection (MAS) has become a valuable tool in plant breeding, especially for developing drought-resistant crop varieties. The identification of genomic regions associated with drought tolerance enables breeders to develop improved cultivars with increased drought tolerance through MAS [90,91]. This approach involves screening for desirable alleles at quantitative trait loci (QTL) for drought tolerance, allowing for the selection of plants with the desired traits [68,92]. MAS has been highlighted as a crucial technique for introgressing known QTLs or genes into elite crop lines to enhance drought tolerance [93]. Recent advancements in crop genomics have facilitated the use of MAS by providing tools to identify and introgress genomic regions associated with drought tolerance, leading to the development of improved cultivars with enhanced drought tolerance [94]. Molecular breeding techniques, such as QTL discovery and marker-assisted wide selection, have shown promise in significantly improving drought resistance in crops [95]. Additionally, the integration of molecular breeding approaches like marker-assisted backcrossing and recurrent selection into crop improvement strategies has been recommended to develop drought-tolerant cultivars, especially in the face of a changing climate [96].

Furthermore, the application of MAS in evolving drought-resistant genotypes is still in an experimental stage, with ongoing efforts focused on identifying markers associated with key drought tolerance traits [64]. The use of various MAS strategies, including marker-assisted introgression, backcrossing, recurrent selection, and gene pyramiding, offers avenues to transfer or introgress desirable traits into crops to enhance drought tolerance [97]. So, MAS plays a pivotal role in breeding drought-resistant crop varieties by enabling the selection of plants with enhanced drought tolerance traits through the identification and

introgression of genomic regions associated with drought tolerance. Integrating MAS with other molecular breeding techniques offers a comprehensive approach to developing cultivars that can thrive under water-deficient conditions, contributing to food security and sustainable agriculture.

3.4. Incorporation of traditional breeding methods with modern biotechnologies

Incorporating traditional breeding methods with modern biotechnologies is crucial for developing water-efficient crops that can thrive under drought conditions. Traditional breeding has been effective in enhancing crop resilience to stress, but the growing global population necessitates the integration of modern biotechnologies to further improve food production [98]. By combining conventional breeding with cutting-edge biotechnological innovations, a robust approach can be established to biofortify major crops, addressing hidden hunger and enhancing global food security. Biofortification, achieved through traditional breeding practices or modern biotechnology, is a promising strategy to increase the nutrient content of crops and combat micronutrient deficiencies [99,100]. Modern biotechnologies offer tools such as marker-assisted selection, transgenic technologies, and "omics" approaches that provide a deeper understanding of drought tolerance mechanisms and aid in identifying candidate genes for breeding drought-tolerant crops [64]. Additionally, the integration of molecular breeding and genetic engineering with conventional breeding can lead to the development of crops that are more resilient to abiotic stresses, including water scarcity [101]. Furthermore, the utilization of advanced biotechnological techniques like genetic engineering and genome editing can accelerate the rate of crop improvement, particularly when combined with speed breeding and high-throughput genotyping [102,103]. To enhance water productivity and resilience in crops, genomics-assisted breeding has emerged as a powerful approach, offering improved plant growth and stress tolerance [104].

The application of biotechnology in breeding can contribute to climate change adaptation and mitigation by increasing agricultural yields and ensuring food security [105]. Moreover, the genetic improvement of oilseed crops through modern biotechnology plays a vital role in enhancing crop productivity, increasing yields, and ultimately securing food availability [106]. The synergy between traditional breeding methods and modern biotechnologies is essential for developing water-efficient crops that can withstand drought conditions, combat hidden hunger, and ensure global food security. By leveraging the strengths of both approaches, researchers and breeders can create crop varieties that are not only high yielding but also resilient to environmental challenges, contributing to sustainable agriculture and improved livelihoods. It is imperative to develop crops that are drought-tolerant and water-efficient due to the effects of climate change and growing water scarcity. This is accomplished by utilizing the variety of methods discussed in Table 2, and these are the most successful at the field level.

Table 2
Methods for creating water-efficient and drought-tolerant crops.

Strategy	Description	Benefits	Challenges	Limitations	Reference
Breeding to resist drought	Choosing cultivars using conventional breeding techniques that are more resilient to heat or drought.	Enhanced crop productivity under adverse conditions which sustains farmer income & global food security	Complex process considering multiple abiotic stresses (high temperatures, high irradiance, nutrient imbalances).	May require long breeding cycles to achieve desired traits.	[57,59]
Developing drought-tolerant cultivars	Characterizing plant physiology to identify effective drought tolerance mechanisms. This can involve analyzing root systems, wilting index, and morphological traits.	Offers farmers a relatively inexpensive crop production approach.	Ongoing efforts to identify reliable markers and quantitative trait loci (QTLs) associated with drought tolerance.	Identifying and selecting for specific traits may not be effective for all environments.	[62,87]
Selecting drought-tolerant cultivars	Prioritizing cultivars that maintain yields under water-deficient conditions. Selection criteria often focus on specific root and morphological traits like root depth and density	Increased water uses efficiency (WUE) in crops.	Requires identifying and selecting for specific traits that may vary depending on the crop species.	Trait selection may not be effective in the presence of other stresses.	[63,66,67]
Marker-assisted selection (MAS)	Utilizing molecular markers linked to drought tolerance genes to select breeding lines	Faster breeding cycles compared to traditional methods. - Increased selection accuracy for desired traits.	Requires significant upfront investment in marker development. - May not be effective for complex traits influenced by multiple genes.	MAS relies on identifying DNA markers associated with drought tolerance, allowing for targeted selection of desirable traits in breeding programs.	[90,91]
Genetic engineering for drought tolerance	Manipulating genes to introduce novel drought tolerance mechanisms.	Potential for creating highly drought-tolerant crops.	Ethical concerns and regulatory hurdles.	Potential unintended consequences.	[79]
Water use efficiency (WUE) breeding	Breeding for cultivars with improved water use efficiency, allowing them to produce more biomass with less water.	Reduces water consumption for agricultural purposes and contributes to sustainable water management practices.	Complex genetic underpinnings of WUE	Difficulty in accurately measuring WUE in field conditions.	[69,72]
Root phenotyping	Utilizing techniques to assess root architecture and function, aiding in selection for drought-tolerant genotypes.	Provides valuable insights into root behavior under water stress conditions.	Limited availability of high-throughput and non-invasive root phenotyping methods.	Root phenotyping methods may not always translate to improved drought tolerance in the field.	[67]
Management practices for water conservation	Implementing irrigation techniques and soil management strategies that optimize water use efficiency.	Reduces water use for agriculture while maintaining or improving yields.	Requires specific knowledge and expertise for implementation.	Effectiveness of management practices can vary depending on soil type, climate, and crop species.	[88]

4. Soil moisture monitoring and management

4.1. Use of remote sensing technologies for mapping soil moisture content

An essential component of the hydrological system, soil moisture significantly impacts the complex interactions between water and energy transfers between the atmosphere and the ground surface. Advances in satellite remote sensing technologies have led to the development of numerous approaches for tracking soil moisture, each with unique benefits and drawbacks [107]. Fig. 3 shows the classifications, including the method of remote sensing technologies that are used in the measurement of soil moisture, and Fig. 4 illustrates their types.

In agriculture, remote sensing techniques, including spectral reflectance analysis, thermal infrared imaging, and radar remote sensing, are increasingly utilized for gathering comprehensive data on soil health, moisture, erosion, and composition over vast areas [108]. Numerous studies have proposed various techniques to extract soil moisture data from remote sensing data types essential for disaster forecasting, environmental surveillance, and hydrological applications [109]. There are two main types of soil moisture monitoring: direct and indirect monitoring [110]. Direct methods for measuring soil moisture involve direct sampling, but these are destructive and slow. Indirect methods offer a faster and less disruptive approach, but their accuracy can be influenced by soil characteristics and sensor limitations. Ultimately, the

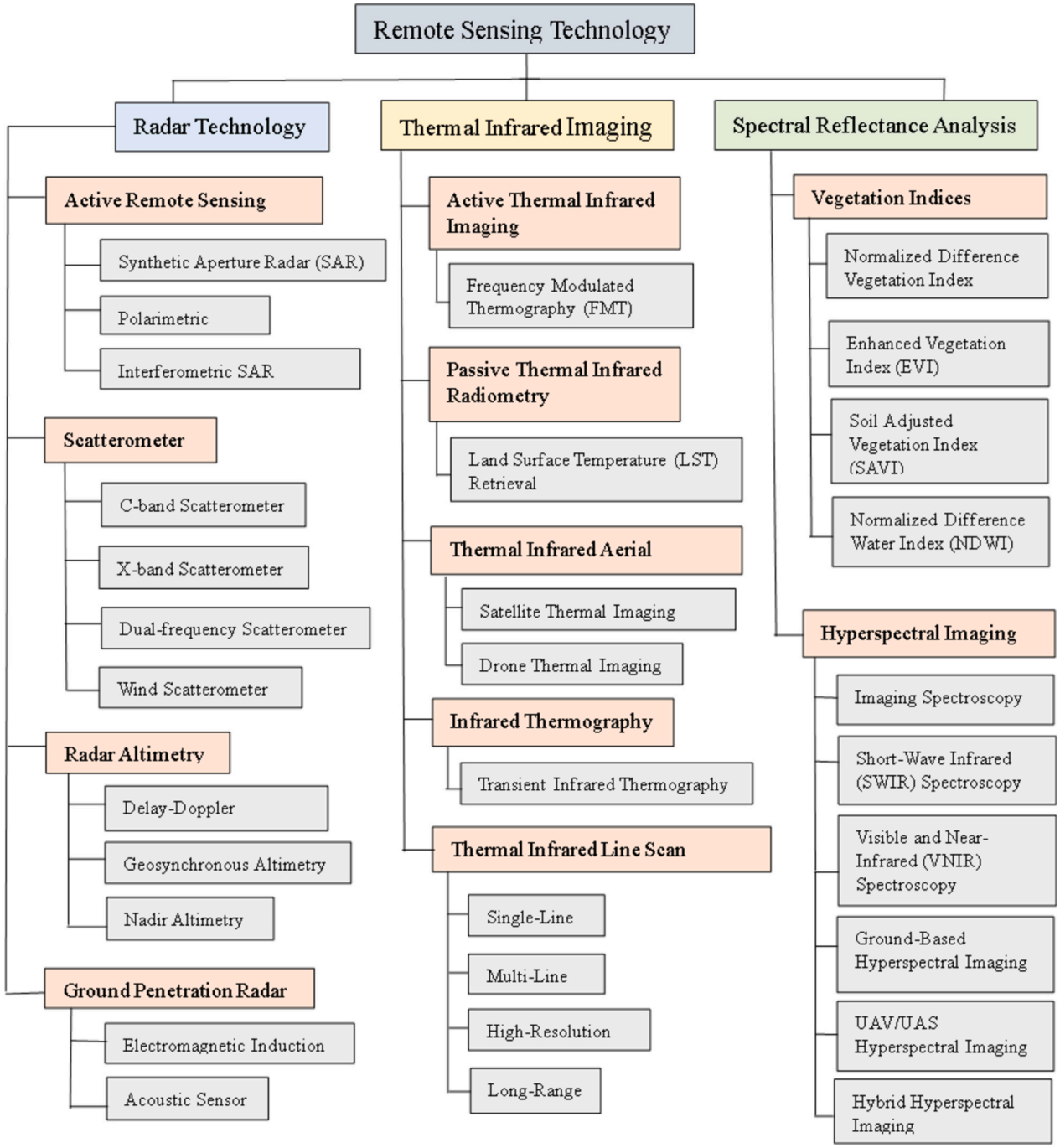


Fig. 3. Types and methods of remote sensing technology in soil moisture measurements
NOTE: This figure covers most of the types and methods of remote sensing technology in soil moisture measurements. UAV = unmanned aerial vehicle; UAS = unmanned aircraft system.

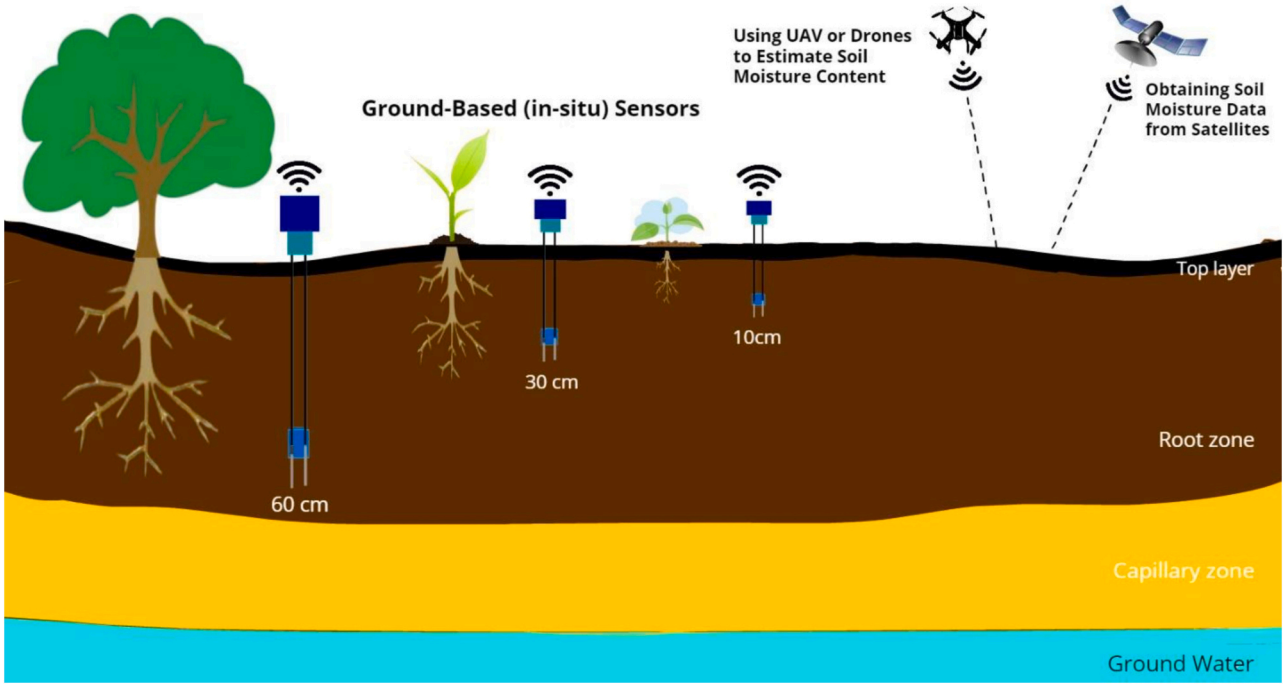


Fig. 4. Different Types of Sensors and Remote Sensing Technology to Measure Soil Moisture.

effectiveness of a soil moisture sensor depends on both the technology and the specific soil environment.

4.2. *Installation of soil moisture sensors for real-time monitoring*

Effective agricultural management requires the installation of soil moisture sensors for real-time monitoring. While satellite and drone-based remote sensing offer larger coverage, in-situ sensors only provide localized data in Fig. 4. For accurate in-situ measurements, representative site selection, careful sensor selection, and appropriate installation are necessary. The stages of gathering, preparing, retrieving, and validating data are essential to utilizing satellite data to monitor soil moisture. Soil moisture estimates can be obtained in real-time and accurately using drones fitted with specialized sensors.

4.2.1. *Ground-based (in-situ) sensor installation*

A number of factors must be carefully taken into account for the effective installation of in-situ soil moisture sensors for real-time monitoring. The selection of sensors ought to be guided by the intended use, suitability for the data recording system, and the properties of the soil (e.g., tensiometers for heavier clays, capacitance for mineral soils) [111]. Once selected, it is essential to find a representative spot in the field that depicts typical soil characteristics. It is best to stay away from areas close to streams or disturbances. To enable real-time data transfer, it is also necessary to ascertain data logger compatibility and communication techniques (wired or wireless). There are multiple crucial steps in the installation procedure itself. For TDM sensors, the suggested depth to insert the sensor is usually between 10 and 60 cm, which is inside the target plant's root zone [112]. Making a clean access hole only slightly wider than the sensor requires less earth disturbance when done using a soil auger. Good soil contact and the removal of air gaps are ensured by a precise and straight insertion into the hole's sidewall. Certain types of sensors may need to be pre-soaked before being installed. The removed soil must be backfilled into the empty gap for optimal contact. To ensure correct data gathering, sensor cables are connected to the data logger according to the instructions given by the manufacturer. The precision of the readings can be further improved by optional calibration. This requires collecting representative soil samples from the installation site, analyzing them in a lab to ascertain their gravimetric water content, and then utilizing the data logger software to adjust the sensor values [113].

4.2.2. *Obtaining soil moisture data from satellites*

Though highly localized data is provided by in-situ sensors, satellite-based remote sensing is a useful tool for detecting fluctuations in soil moisture at the field size. Data accuracy can be impacted by various factors such as sensor type, revisit frequency, geometrical problems, spectral problems, mapping limits and challenges, and atmospheric conditions [114]. There are several steps involved in using satellite data for soil moisture monitoring. First, based on their research objectives, researchers select a particular satellite mission and its data product (e.g., SMAP, ESA CCI) for satellite data collection [115]. This decision is influenced by variables such as the desired geographical (area covered) and temporal (frequency of data collection) resolution. Data preprocessing is done on the collected data. Here, corrections are made for geometric distortions (accounting for Earth's curvature and sensor position), atmospheric effects [116], and radiometric calibration [117] (compensating for equipment fluctuations). This process raises the data's overall quality and accuracy. Next, soil moisture retrieval is applied to the preprocessed data. Estimates of the true soil moisture content are produced from the satellite measurements using specialized algorithms, which may include brightness temperature and vegetation indicators [118]. Lastly, researchers carry out data validation to verify the accuracy of these estimations. This includes comparing the soil moisture determined by satellite with readings from in-situ sensors or other known methods [119]. This verification process helps in determining how trustworthy satellite data is for tracking soil moisture.

4.2.3. *Using UAV or drones to estimate soil moisture content*

Although satellite data delivers significant insights at a regional level, drone technology presents an additional powerful instrument for tracking soil moisture. More effective monitoring is possible due to satellite data's greater spatial resolution and shorter temporal resolution. In addition, remote sensing has advanced quickly since the introduction of unmanned aerial vehicles (UAVs), enabling highly precise and real-time monitoring [120]. By identifying the electrical

characteristics of the soil, drones fitted with specialized sensors—like radar or microwave sensors—can determine the moisture content of the soil. Because of their sensitivity to variations in soil moisture, these sensors make it possible to estimate moisture levels precisely from 10 to 20 cm of the soil profile [121].

4.3. Implementation of conservation tillage practices to enhance soil water retention

Crop production with profitability in agriculture in semi-arid and arid regions is highly variable each year, and these two variables are directly influenced by the amount and distribution of rainfall that occurs throughout the growing season. Conservation tillage can help conserve water and increase crop water usage efficiency. Instead of turning the soil over (traditional/conventional tillage), conservation tillage reduces disturbance and leaves crop residue on the surface. A "no-till" strategy keeps moisture and inhibits erosion by acting as a blanket. At first, conventional tillage may seem beneficial, but it loosens the soil, which allows valuable moisture to escape like a leaky bucket. Therefore, conservation tillage, which combines minimum and zero tillage, offers to increase crop production while having the least negative environmental impact. It also has the potential to break the surface compact zone in soil with less soil disturbance. So, conservation tillage is better than conventional tillage. The terms stubble mulching, eco-fallow, minimum tillage, reduced tillage, no-tillage, and direct drill are often used to describe conservation tillage. These plant residue management systems are achieving threefold objective: lowering energy consumption, saving soil and water, and maintaining an adequate amount of plant residue on the soil surface for controlling wind and water erosion [122].

4.4. Adoption of cover cropping and mulching techniques for soil moisture conservation

Crops that are not harvested but are grown to benefit the soil and/or other crops in a number of ways are called cover crops [123]. Cover crops are excellent for maintaining healthy soil, reducing soil erosion, preserving nutrients, and fixing nitrogen; they also compete with subsequent crops for water resources. There may be two sides to this effect on soil moisture. Proper management of cover crops can enhance water infiltration, decrease evaporation, and even aid in drying out excessively damp soil to facilitate quicker planting. However, if not managed carefully, they might worsen moisture levels or exhaust the water required for the subsequent crop [124]. Mulching is the covering of plants with a layer of organic material, such as leaves, straw, or peat, to keep moisture from evaporating, which also minimizes weed infestation, controls water evaporation, and lessens soil deterioration by reducing runoff and loss of soil. Consequently, it facilitates the retention of soil moisture, helps control temperature, improves the physical, chemical, and biological properties of soil, enriches soil with nutrients, and ultimately accelerates crop growth and yield. Cover crops have many benefits for soil health, but their water consumption needs to be managed carefully to avoid depleting resources for later plantings. Fortunately, regular mulching is an effective way to retain soil moisture, promoting sustainable land management practices and agricultural productivity. Proper irrigation techniques, along with cover cropping, can also help reduce the risk of water depletion and ensure optimal soil moisture levels for subsequent crops. By using an integrated approach, agriculture can benefit from cover crops without compromising the necessary water requirements of future harvests.

5. Nanotechnology applications for water use efficiency

5.1. Nano-sensors for monitoring soil moisture and nutrient levels

5.1.1. Principles of nano-sensor technology

Nano-scale sensors are essential in transforming soil monitoring practices, especially in evaluating soil moisture and nutrient levels to optimize water usage in agriculture. These sensors, known for their small size and high sensitivity, allow precise and real-time monitoring of soil conditions [125]. Through the utilization of wireless sensor networks, nano-scale sensors can offer high temporal resolution data at various spatial scales, providing a detailed insight into soil moisture dynamics [126]. The use of wireless sensor networks can also enable near-real-time monitoring of soil water content variability, thereby improving water utilization efficiency in agricultural practices in Fig. 5 [127–129].

Advancements in sensor technologies, such as the development of low-cost capacitive-based soil moisture sensors, have facilitated the creation of smart monitoring units powered by solar cells for managing greenhouse irrigation [130]. Strategically placing these sensors in the soil allows for accurate and site-specific measurements of soil moisture levels, aiding in precise water resource management in agriculture [131]. Furthermore, integrating multiple sensors and fostering collaboration between different sensor types have been demonstrated to enhance regional soil moisture monitoring capabilities, thereby enhancing the quality and timeliness of data collection [118]. Nano-scale sensors, when combined with wireless sensor networks and advanced monitoring technologies, serve as a potent tool for monitoring soil moisture and nutrient levels with high precision. By enabling real-time data collection and analysis, these sensors contribute significantly to improving water utilization efficiency in agriculture, ultimately leading to enhanced crop yields and sustainable farming practices.

5.1.2. Types of nano-sensors

Nanosensors play a crucial role in monitoring soil moisture and nutrient levels in agriculture. These sensors come in various types, each with unique capabilities and applications. Capacitive soil moisture sensors, such as the ECH2O probe models, have been evaluated for their sensitivity to factors like supply voltage, temperature, and soil electrical conductivity, providing accurate readings for soil moisture levels [132]. Additionally, low-cost capacitive and resistive soil moisture sensors have been calibrated and compared, with capacitive sensors showing higher accuracy and precision, making them suitable for precision agriculture applications [133]. In the realm of precision agriculture, miniature integrated multimodal sensors have been developed to measure pH, electrical conductivity (EC), and temperature, offering a comprehensive approach to monitoring soil conditions for optimal crop growth [134]. These sensors provide essential data for efficient irrigation water management and greenhouse operations, contributing to sustainable agricultural practices [130,135]. Moreover, the development of smart monitoring systems utilizing IoT devices like Particle Electron or Particle Proton Arduino-compatible devices enables real-time data communication for soil moisture monitoring [136]. Furthermore, advancements in nanotechnology have led to the creation of nanosensors that respond to environmental conditions, converting them into valuable information for sustainable agriculture practices [125]. These nano-sensors, along with IoT-based systems, offer innovative solutions for monitoring soil parameters and optimizing irrigation practices in agriculture [137]. Additionally, the integration of nanosensors with IoT technologies allows for smart monitoring of soil conditions, enhancing the efficiency and productivity of agricultural operations [137]. So, the diverse array of nanosensors available for monitoring soil moisture and nutrient levels in agriculture provides farmers and researchers with valuable tools to optimize irrigation practices, enhance crop productivity, and promote sustainable agricultural practices.

5.1.3. Advantages and limitations

Monitoring soil moisture and nutrient levels using nanosensors offers numerous benefits, including high sensitivity and real-time monitoring capabilities. Nanosensors provide precise and accurate data, enabling farmers to make informed decisions promptly. These sensors can detect

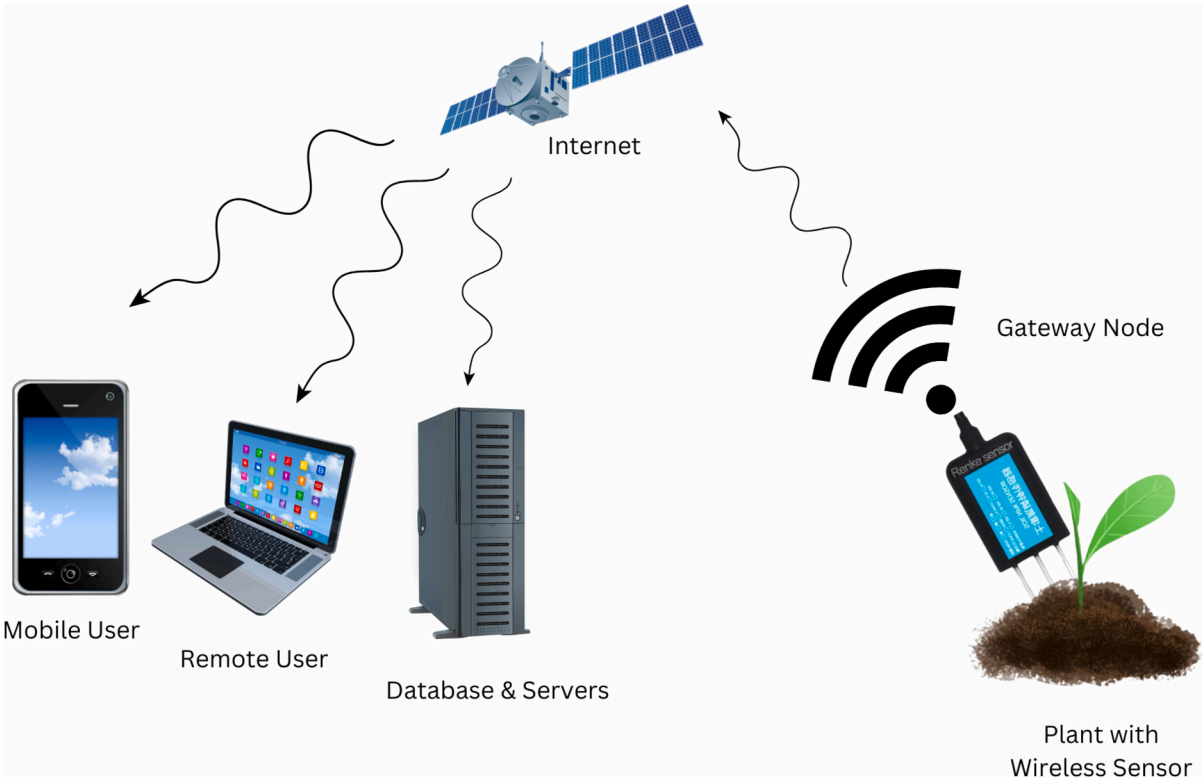


Fig. 5. Nano-sensors for monitoring soil moisture and nutrient levels.

subtle changes in soil conditions, allowing for timely adjustments in irrigation and fertilization practices to optimize crop growth and yield [125,138]. Real-time monitoring facilitated by nanosensors enhances the efficiency of agricultural operations by providing immediate feedback on soil conditions. This capability enables farmers to respond promptly to changes in moisture and nutrient levels, ensuring optimal growing conditions for crops. Additionally, the high sensitivity of nanosensors allows for the detection of minor variations in soil parameters, leading to improved resource management and increased agricultural productivity [133,139,140]. However, challenges such as cost and calibration need to be addressed when utilizing nanosensors for soil monitoring. The initial investment in acquiring nanosensor technology can be a barrier for some farmers, especially smallholders. Additionally, the calibration of nanosensors is crucial to ensure accurate and reliable data collection. Proper calibration procedures are essential to account for environmental factors that may influence sensor readings, such as temperature and soil composition. So, the use of nanosensors for monitoring soil moisture and nutrient levels offers significant advantages in terms of high sensitivity and real-time monitoring capabilities. While challenges related to cost and calibration exist, addressing these issues can unlock the full potential of nano-sensor technology in agriculture, leading to improved sustainability and productivity in farming practices.

5.2. Nano-fertilizers and soil amendments for water retention and nutrient uptake

5.2.1. Composition and mechanisms of nano-fertilizers

Nano-fertilizers are a promising innovation in modern agriculture due to their potential to enhance nutrient uptake efficiency, reduce environmental impact, and improve crop yields. These nanostructured materials offer unique properties, such as a high specific surface area, which can facilitate the controlled release of nutrients to plants [141]. The design and composition of nano-fertilizers involve considerations such as the size, shape, chemical composition, porosity, and surface charge of nanoparticles to optimize their performance [142]. Researchers have explored various approaches to developing nano-fertilizers with controlled release mechanisms. For instance, incorporating NPK fertilizer into chitosan nanoparticles has been investigated, although there is still room for further exploration in utilizing chitosan nanoparticles for controlled release of nutrients [143]. Additionally, studies have focused on the synthesis and characterization of zeolite-based nanocomposites as slow-release fertilizers [144], highlighting the potential of environmentally friendly nano-fertilizers in Fig. 6.

Nano-fertilizers have been shown to enhance nutrient utilization and gene expression in crops, leading to improved nitrogen utilization and overall plant growth [145]. Furthermore, the use of nanotechnology in agriculture offers the opportunity to design novel materials that are more readily absorbed by plant roots, thereby enhancing nutrient use efficiency [146]. Nano-fertilizers can also be tailored for specific applications, such as foliar application, to alleviate stress conditions in crops and promote growth and yield [147]. The controlled release of nutrients from nano-fertilizers is crucial for achieving sustainable and environmentally sound agriculture. By precisely regulating nutrient release, nano-fertilizers can help optimize plant growth, reduce chemical pollution, and increase agricultural productivity [148]. The development of nano-fertilizers with slow-release properties opens up new avenues for precision farming and sustainable agricultural practices [149]. The design and composition of nano-fertilizers with controlled release mechanisms hold great potential for revolutionizing modern agriculture by improving nutrient efficiency, reducing environmental impact, and enhancing crop yields. Further research and development in this field are essential to unlocking the full benefits of nano-fertilizers in sustainable agricultural practices.

5.2.2. Soil amendments with nano materials

Nano-materials have demonstrated potential in enhancing soil structure, water retention, and nutrient availability in agricultural systems. The unique properties of nano-materials, such as their high specific surface area and chemical reactivity, play a crucial role in

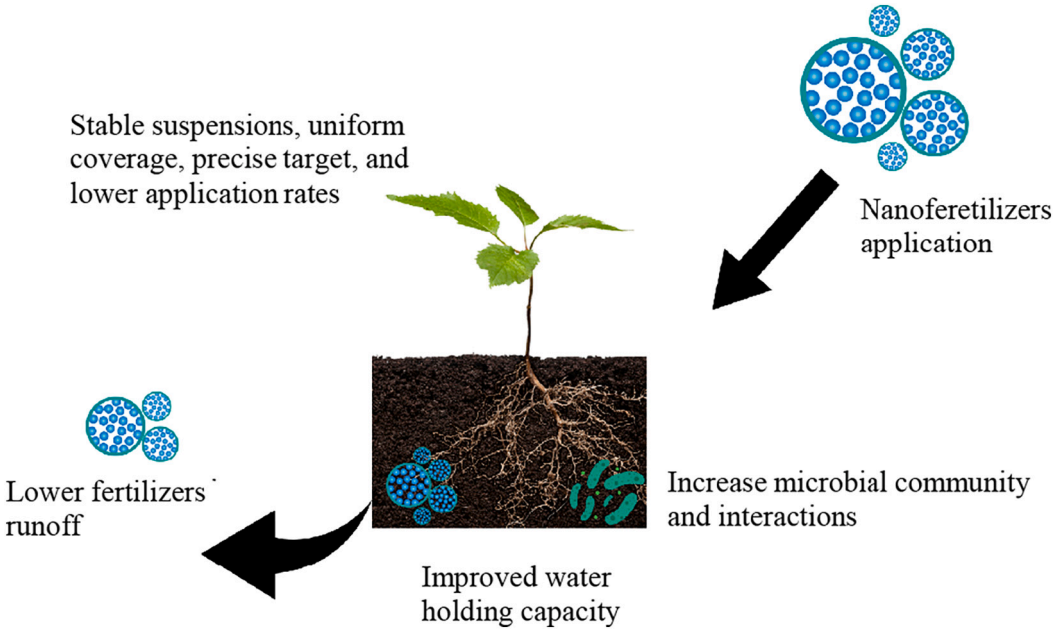


Fig. 6. Application of nano-fertilizers and soil amendments.

improving soil quality and nutrient management [150]. Research has shown that the use of nano-materials as nano-fertilizers can synergistically increase soil nutrient availability and enhance crop production [150]. For example, the application of nano-biochar and copper oxide nanoparticles has been found to improve soil nutrient availability and increase wheat production [150]. Moreover, incorporating nano-materials like carbon nanoparticles into soil has been reported to enhance soil fertility by increasing the storage capacity of soils for available nutrients, such as ammonium nitrogen and available phosphorus [151]. This enhancement is attributed to the large specific surface area of carbon nanoparticles, which can retain and complex nutrients in the soil, thereby improving nutrient availability for plant uptake [151]. Additionally, the utilization of magnetic nanoparticles grafted with poly (acrylic acid) has been explored to enhance soil nutrient adsorption capacity and water holding capacity, contributing to improved nutrient availability in agriculture [152]. Furthermore, studies have investigated the role of nano-materials in enhancing microbial activity in the soil, crucial for nutrient cycling and availability. The application of silica nanoparticles has been shown to increase the population of phosphate-solubilizing bacteria in soil, leading to enhanced soil fertility and improved nutrient availability for plants [153]. Similarly, the mobilization of native phosphorus in the rhizosphere using zinc oxide nanoparticles synthesized by soil fungi has been found to increase soil phosphorus availability and promote plant growth [154]. So, the application of nano-materials in agriculture presents a promising approach to improving soil structure, water retention, and nutrient availability. By leveraging the unique properties of nano-materials, such as their high surface area and reactivity, farmers can enhance soil fertility, promote nutrient cycling, and ultimately increase crop productivity sustainably.

5.2.3. Field applications and performance

To assess the effectiveness of nano-fertilizers and soil amendments in field trials, various studies provide valuable insights into their impact on soil quality, nutrient availability, and crop productivity. One study by Seliman et al. [155] discusses the potential benefits of nano-fertilization in modern agriculture, highlighting the advantages of using nano-fertilizers to improve nutrient management and reduce environmental nutrient losses [155]. Additionally, it investigates the response of wheat cultivars to different fertilizers, including bio- and nano-fertilizers, emphasizing the positive effects of these amendments on wheat crop growth [156]. These studies demonstrate the potential of nano-fertilizers in enhancing crop productivity in field trials. Regarding soil amendments, research by Ch'Ng [157] explores the use of biochar and compost derived from agro-industrial wastes to improve soil phosphorus availability and enhance the yield of *Zea mays* L. in field trials [157]. Similarly, Sharma [158] evaluated the survival of *Escherichia coli* in manure-amended soils under different spatiotemporal and agricultural conditions, highlighting the importance of field studies in understanding the impact of amendments on soil microbiology [158].

Furthermore, the effectiveness of biochar amendments in suppressing phytophthora blight in sweet pepper crops is investigated by Shoaf [159], emphasizing the potential of biochar to mitigate plant diseases in field trials. Additionally, Kumar [160] focus on the impact of organic manure, bio-fertilizers, and nano-fertilizers on crop yield and soil nutrient status, highlighting the enhanced effectiveness of nano-fertilizers in improving nutrient uptake by crops [160]. Overall, these studies collectively demonstrate the positive effects of nano-fertilizers and soil amendments in field trials, showcasing their potential to enhance soil quality, nutrient availability, and crop productivity in agricultural systems.

5.3. Nanomaterial-based membranes for water filtration and desalination

Nanomaterial-based membranes have significantly enhanced water filtration processes, particularly in desalination and water purification applications. Membranes incorporating artificial water channels have demonstrated notable improvements in efficiency for brackish and low-salinity water reverse osmosis desalination [161]. The utilization of nano-materials in membranes has indeed enhanced filtration efficiency, enabling the effective removal of contaminants from water sources like brackish or saline water [162,163]. The development of membranes with tunable properties has provided better control over the filtration process, resulting in higher performance in removing impurities from water [161]. Nanomaterial-based membranes have played a crucial role in improving water reuse practices, particularly in irrigation. Membranes such as nano-filtration and brackish reverse osmosis membranes have been pivotal in enhancing the removal efficiency of target contaminants, thereby making water suitable for irrigation purposes [163]. The integration of nanotechnology into membrane design has optimized

the selectivity and permeability of membranes, facilitating the treatment of water from various sources, including brackish or saline water, for irrigation reuse [161]. Nanomaterial-based membranes have made significant contributions to advancing water filtration efficiency, enabling the reuse of irrigation water and the utilization of brackish or saline water sources. These membranes have exhibited superior performance in removing contaminants and enhancing water quality, establishing their importance in sustainable water management practices.

5.4. Potential environmental implications and safety concerns of nanotechnology in agriculture

5.4.1. Assessing environmental impact of nanoparticles

The release of nanoparticles into the environment poses significant risks across various ecosystems. Nanoparticles, due to their high reactivity and physiochemical dynamics, can interact with living organisms and environmental media, leading to potential adverse effects [164]. When nanoparticles are released into natural ecosystems, they undergo dynamic interactions with organisms, altering their chemical and physical properties [165]. These interactions can result in the formation of protein coronas on the nanoparticle surfaces, affecting their behavior and impact on living systems [166,167].

Studies have shown that the release of engineered nanoparticles into the environment can lead to concerns about their potential phytotoxic effects on plants [168]. Furthermore, the presence of nanoparticles in water sources can impact plant growth and food standards, highlighting the potential risks associated with nanoparticle contamination in irrigation water [169]. Nanoparticles, such as titanium dioxide, have been found to be released into the environment from various sources, including paints, and can persist in urban runoff, contributing to environmental contamination [170]. The environmental fate of nanoparticles, such as cerium nanoparticles, can be influenced by the exposure scenario, with chronic low-level releases from industrial and consumer products contributing to their presence in aquatic ecosystems [171]. Additionally, the genotoxic impacts of silver nanoparticles raise concerns about their use in household water treatment, emphasizing the need for careful consideration of their potential risks [172]. So, the release of nanoparticles into the environment can have far-reaching consequences on soil health, water quality, and ecosystem dynamics. Understanding the interactions of nanoparticles with living organisms and environmental media is crucial for assessing and mitigating the risks associated with nanoparticle contamination in various ecosystems.

5.4.2. Mitigating environmental risks

To mitigate potential environmental hazards associated with the release of nanoparticles, various strategies can be implemented. One effective approach is the development of nanoparticle containment technologies to prevent their uncontrolled release into the environment. Containment technologies can include the use of specialized coatings or encapsulation methods to ensure that nanoparticles remain securely bound and do not leach into surrounding ecosystems [173,174]. By implementing containment measures during the production, use, and disposal of nanoparticles, the risk of environmental exposure can be significantly reduced. Additionally, environmentally friendly disposal methods play a crucial role in minimizing the impact of nanoparticles on soil health, water quality, and ecosystem dynamics. Strategies such as recycling and proper waste management can help prevent the accumulation of nanoparticles in the environment [175]. Developing disposal methods that focus on the safe handling and treatment of nanoparticle-containing waste can prevent their dispersion into natural systems [139]. Furthermore, the use of advanced materials in disposal processes, such as nano-materials that can efficiently extract metal ions, can aid in reducing secondary pollution and operational costs. Moreover, the application of nanotechnology in wastewater treatment systems can help address the challenges posed by nanoparticle contamination. Engineered nanoparticles can be utilized to enhance the efficiency of wastewater treatment processes, ensuring the removal of hazardous pathogens and microorganisms before discharge into the environment. By incorporating nano-materials into treatment systems, the risks associated with nanoparticle release can be mitigated effectively.

5.4.3. Balancing innovation with environmental sustainability

Responsible innovation in nanotechnology is crucial for the development and deployment of nano-enabled agricultural technologies. The integration of nanotechnology in agriculture, such as in the treatment of plant diseases, controlled release of fertilizers, and plant breeding, highlights the importance of a rational design of nano-carriers to ensure environmental sustainability and safety [176]. Prioritizing responsible innovation practices that focus on minimizing risks and maximizing benefits in the agricultural sector is essential. Developing a framework for responsible innovation is vital to guide the scientific community in ensuring the safe and sustainable application of nanotechnology in agriculture [177]. This framework should emphasize adequate risk management strategies for workers, occupational safety practices, and the development of responsible regulatory consensus to address the toxicological implications and occupational risks associated with nanotechnology in agriculture. Defining risk management strategies and regulatory frameworks can help the agricultural sector effectively address the challenges posed by the integration of nanotechnology. Then regulation plays a significant role in promoting responsible innovation in nanotechnology in food and agriculture [178]. Implementing the principles of responsible innovation can assist innovators and policymakers in anticipating risks and addressing societal concerns, thereby fostering the development of safe and sustainable nano-enabled agricultural technologies. Overcoming barriers to responsible innovation, as identified by experts and developers in the field, is crucial for ensuring the ethical and sustainable deployment of nanotechnology applications in food and agriculture [179]. So, responsible innovation in nanotechnology for agriculture necessitates a multidisciplinary approach that considers environmental sustainability, safety, and societal implications. By prioritizing responsible innovation practices, developing regulatory frameworks, and addressing barriers to implementation, the agricultural sector can leverage the potential of nanotechnology while safeguarding the environment and human health.

6. Artificial intelligence and data analytics

6.1. Use of machine learning algorithms for predictive modeling of water demand

6.1.1. Introduction to machine learning in agriculture

Machine learning has the potential to revolutionize water management in agriculture by utilizing advanced technologies to analyze agricultural data. Through the application of machine learning techniques such as deep learning and predictive algorithms, agricultural stakeholders can improve water management practices to optimize irrigation, conserve water resources, and enhance crop yields [180–182]. These technologies facilitate precise monitoring of soil moisture levels, crop water requirements, and weather patterns, enabling data-driven decision-making in irrigation scheduling and water allocation [180,183]. Additionally, machine learning is instrumental in crop yield prediction, disease detection, and soil management, which are crucial aspects of efficient water usage in agriculture [182,184]. Accurate forecasting of crop yields and early identification of potential disease outbreaks empower farmers to implement targeted interventions, thereby reducing water wastage and increasing overall productivity [182,184]. Machine learning algorithms also support soil classification, species management, and weed detection, contributing to improved water efficiency and sustainable agricultural practices [182]. Furthermore, the integration of big data analytics with machine learning approaches provides

valuable insights for enhancing agricultural yield while optimizing water usage [185]. By analyzing extensive agricultural data, including crop performance, environmental factors, and historical trends, machine-learning models can offer actionable recommendations for water-efficient farming practices [185]. This data-driven approach empowers farmers to make informed decisions regarding irrigation strategies, leading to enhanced water conservation and crop sustainability [185].

6.1.2. Types of machine learning algorithms

To accurately predict water demand, a variety of machine learning algorithms can be utilized. These include supervised, unsupervised, semi-supervised, and reinforcement learning algorithms. By analyzing historical data, these algorithms can estimate lost sales, forecast future demand, and help prevent water wastage [186,187]. Statistical and machine learning algorithms are essential for precise water demand prediction, supporting informed decision-making in water resource management [186]. Machine learning techniques such as Artificial Neural Networks (ANN), Support Vector Machines (SVM), Random Forest, and Naïve Bayes have been successfully used to predict water quality features, highlighting the versatility of these methods in water-related applications [188]. Additionally, ensemble machine learning algorithms like eXtreme Gradient Boosting (XGBoost) have been effectively combined with models such as the Soil and Water Assessment Tool (SWAT) to predict river water salinity, showcasing the value of machine learning in improving existing models for water-related predictions [189]. The wide array of machine learning algorithms available, ranging from traditional methods like Random Forest to advanced techniques like XGBoost, offers a robust toolkit for predictive modeling of water demand. By leveraging these algorithms, stakeholders can make well-informed decisions, mitigate water scarcity, and optimize water resource management practices effectively.

6.2. Data-driven decision support systems for optimizing irrigation practices

6.2.1. Development and components of decision support systems

Decision Support Systems (DSS) for optimizing irrigation practices typically consist of several key components that work together to assist users in making informed decisions regarding irrigation management. The architecture of a DSS includes data input mechanisms, processing algorithms, and a user interface to facilitate interaction with the system. Data input in a DSS for irrigation optimization can involve various sources such as sensor data from the field, weather information, soil moisture levels, and crop water requirements [190]. These inputs are crucial for the system to analyze the current conditions and make recommendations for irrigation scheduling. Advanced monitoring systems integrated into DSS can provide real-time data to improve the accuracy of decision-making [190]. The processing algorithms within a DSS are designed to analyze the input data and generate meaningful insights for users. These algorithms can range from simple calculations to complex models that consider multiple variables simultaneously. For instance, integrating adaptive decision support systems into precision irrigation management can significantly enhance the efficiency of irrigation practices [190]. Additionally, intelligent DSS can combine human expertise with modeling tools to assist decision-makers in flood management scenarios [191]. The user interface of a DSS plays a crucial role in presenting the processed information to users in a comprehensible manner. A well-designed interface allows users to interact with the system, visualize data, and receive recommendations for irrigation practices. Decision support systems for irrigation scheduling have been developed with flexible interfaces tailored to the needs of users in specific regions, such as in China [107]. So, the architecture and components of a DSS for optimizing irrigation practices involve data input mechanisms to gather relevant information, processing algorithms to analyze the data and generate recommendations, and a user interface to present the information to users. By integrating advanced monitoring systems, intelligent algorithms, and user-friendly interfaces, DSS can effectively support decision-making processes in irrigation management.

6.2.2. Real-time monitoring and feedback

Decision Support Systems (DSS) utilize real-time sensor data, remote sensing imagery, and weather forecasts to provide timely irrigation recommendations. Real-time sensor data, including soil moisture levels and plant temperatures, collected through wireless networks of automated sensors, offers continuous feedback to DSS for recalibrating and adjusting irrigation recommendations [192]. This feedback enables the optimization of irrigation practices based on current field conditions [190,193]. Remote sensing imagery is essential for monitoring crop health and water stress levels, allowing DSS to evaluate the irrigation needs of different areas within a field [194]. By integrating remote sensing data, DSS can offer site-specific irrigation recommendations tailored to the requirements of individual crops [195]. Weather forecasts are another crucial component used by DSS to anticipate upcoming environmental conditions and adjust irrigation schedules accordingly [196]. By incorporating weather predictions into the decision-making process, DSS can proactively manage irrigation to accommodate changing weather patterns and optimize water usage [197]. The integration of real-time sensor data, remote sensing imagery, and weather forecasts into DSS enables farmers to make informed irrigation decisions promptly. By leveraging these technologies, DSS can enhance water efficiency, improve crop productivity, and support sustainable irrigation practices in agriculture.

6.3. Integration of satellite imagery and drone data with AI for precision agriculture

6.3.1. Remote sensing technologies in precision agriculture

Remote sensing technologies, such as satellite imagery and drone technology, are essential in precision agriculture by providing valuable spatial data for monitoring and managing agricultural activities. Satellite remote sensing is widely used for estimating land use, land cover, crop biomass, and chlorophyll content in crops [198]. Drones have gained prominence in agriculture due to their ability to offer high-resolution data through lower flight altitudes, enabling precise monitoring of individual areas [199]. Drones provide farmers with a close-up view of their fields, allowing for more accurate evaluations compared to satellite technology [200]. In precision agriculture, remote sensing technologies are utilized to promptly detect pests, diseases, and weeds, informing farmers of the state of affairs in their fields [201]. Drones are increasingly used for spraying pesticides, further enhancing their utility in various agricultural tasks [202]. The integration of satellite imagery and drone technology allows for the refinement of vegetation indices, aiding in crop mapping, monitoring, and disease assessment [203]. Remote sensing technologies also facilitate the estimation of chlorophyll content in crops, contributing to improved precision agriculture practices [204]. The combination of satellite, UAV, and ground inversion techniques enhances the accuracy of monitoring crop characteristics like chlorophyll content, supporting better decision-making in agriculture [198]. Additionally, the use of drones in conjunction with machine learning has been shown to refine vegetation indices for precision agriculture applications [203]. Overall, the synergy between satellite imagery and drone technology offers a comprehensive approach to collecting spatial data in precision agriculture. These technologies enable farmers to make informed decisions, optimize resource management, and enhance agricultural productivity through precise monitoring and targeted interventions.

6.3.2. AI-driven image analysis techniques in precision agriculture

AI-driven image analysis methods are essential in deriving actionable insights from remote sensing data in precision agriculture. Deep

learning techniques, such as semantic segmentation networks, facilitate the automatic discovery of relevant contextual features for improved image classification results [205]. Transfer learning has been effectively utilized in semantic segmentation to differentiate between crops and weeds, demonstrating excellent classification performance [206]. Anomaly detection methods, including those employing generative adversarial networks with attention mechanisms, show promise in detecting anomalies in IoT time-series data in smart agriculture [207]. Ensemble methods, like the one proposed in the framework for environment perception, amalgamate outputs from various tasks such as semantic segmentation, object detection, and anomaly detection to enhance scene understanding in agriculture [208]. Deep neural networks have been employed in large-scale semantic weed mapping frameworks, utilizing aerial multispectral imaging data for precise weed mapping in precision farming [209]. Furthermore, image-to-image translation-based data augmentation techniques have been used to enhance crop/weed classification models, underscoring the significance of high-quality data in improving classification accuracy in precision agriculture [210]. These AI-based image analysis methods showcase the potential for advanced data processing and interpretation in precision agriculture, empowering farmers to make informed decisions and optimize agricultural practices based on actionable insights derived from remote sensing data.

6.3.3. Scaling up precision agriculture

To scale up AI-driven precision agriculture practices from individual fields to larger agricultural landscapes, various strategies can be implemented. Firstly, the integration of decision support systems (DSS) in precision agriculture can aid in managing larger areas efficiently by providing real-time data analysis and recommendations [211]. These systems can assist in optimizing resource allocation, enhancing crop management practices, and improving overall sustainability in agriculture [211,212]. Furthermore, leveraging advanced technologies like drones can enable the collection of high-resolution data over extensive areas, facilitating precise monitoring and management of crops at a larger scale [213]. Drones equipped with AI-based image analysis methods can enhance pest management, crop monitoring, and anomaly detection across vast agricultural landscapes [213]. The adoption of blockchain technology in conjunction with the Internet of Things (IoT) can also play a crucial role in scaling up precision agriculture practices. Blockchain can ensure data integrity and traceability, while IoT devices can provide real-time data on various agricultural parameters, enabling informed decision-making at a larger scale [214]. So, to promote sustainable intensification in precision agriculture through the development of environment perception frameworks can contribute to scaling up AI-driven practices [211]. These frameworks can amalgamate outputs from different tasks like semantic segmentation, object detection, and anomaly detection to enhance scene understanding and decision-making in agriculture [211]. Challenges in scaling up AI-driven precision agriculture practices include the need for large-scale software development to meet stakeholder requirements, ensuring data security and privacy in IoT applications, and addressing the economic trade-offs between conservation and crop production [215,216]. However, opportunities lie in increasing agricultural productivity, enhancing sustainability, and improving resource management through the widespread adoption of AI-driven technologies in precision agriculture [216,217]. Overall, by embracing advanced technologies, integrating decision support systems, and promoting sustainable practices, scaling up AI-driven precision agriculture practices from individual fields to larger agricultural landscapes can lead to improved efficiency, productivity, and sustainability in modern agriculture.

6.4. Challenges and opportunities in harnessing AI for water use efficiency in agriculture

6.4.1. Data quality and availability

To address challenges related to data quality, reliability, and accessibility for training AI models in agricultural settings, it is crucial to consider various aspects highlighted in the literature. Ensuring the reliability and relevance of agricultural data is essential [218]. New technological developments have enabled the use of data to prevent crop problems and enhance crop diagnosis accuracy [219]. Ethical considerations, such as transparency, accountability, and sustainability, play a significant role in the responsible use of AI in digital agriculture [220]. Digital twins and real-time data analysis driven by AI have the potential to revolutionize farming practices, leading to better decision-making, improved animal health, and optimized resource utilization [221]. The adoption of AI technology in the agro-food system can enhance transparency, accountability, and sustainability throughout the supply chain [222]. Machine learning algorithms applied to sensor data analytics offer valuable insights within the agricultural ecosystem [223]. Smart agriculture initiatives, including pre-trained models with optimal environmental settings, can boost agricultural product yield and quality [224]. Overcoming challenges related to data sets is crucial for the successful implementation of AI and deep learning in agriculture [225]. Leveraging AI models for on-farm sorting can automate harvesting processes and reduce labor costs [226]. Furthermore, the availability of relevant data is pivotal for AI to deliver value in poverty reduction and various sectors, including agriculture [227]. AI applications in agriculture can lead to economic development, improved farmers' income, and regional growth [228]. Understanding the behavioral factors influencing the acceptance of AI systems in agriculture is essential for successful implementation [229].

6.4.2. Technological adoption and adaptation

To address the adoption barriers and opportunities for integrating AI technologies into existing agricultural practices, several factors need consideration. The adoption of AI in agriculture faces challenges such as lack of financial resources, need for special training, performance risk, perceived cost, technology dependency, need for human interaction, and fear of AI replacing human work [230]. Additionally, barriers like lack of connection between institutions, weak relationships, and diverging interests hinder technology adoption in agriculture [231]. On the other hand, opportunities exist for AI integration in agriculture, such as the potential to optimize production systems to feed a growing global population while conserving agricultural land and preserving environmental quality [232]. Timely and reliable technological information flow can decrease uncertainty in technology adoption and advance the adoption of agricultural technology [233,234]. Moreover, the adoption of AI technologies in agriculture can lead to sustainable business models that protect consumers and improve the quality of agricultural production [222]. Factors influencing the adoption of agricultural technology, including AI, are multifaceted and depend on personal, social, cultural, and economic aspects, as well as the characteristics of the innovation itself [235]. Capacity-building activities can help in formulating adaptation measures, increasing productivity, and improving the capacity to adapt to climate change pressures, thereby facilitating the adoption of climate-smart agriculture using intelligent techniques [236].

6.4.3. Policy and regulatory considerations

To promote responsible AI deployment in agriculture, policies and regulations play a crucial role in ensuring data governance, privacy protection, and algorithm transparency. The responsible use of AI in agriculture requires appropriate policy frameworks to govern the deployment of AI technologies [220]. These frameworks should address ethical, regulatory, and practical considerations to enable widespread and scalable use of AI in the agricultural sector [237]. Policies need to

focus on data governance to ensure that data collected and utilized by AI systems are handled ethically and securely. Additionally, regulations should emphasize privacy protection to safeguard farmers' data and prevent unauthorized access or misuse [238]. Algorithm transparency is another key aspect that policies should address to enhance trust in AI systems [239]. Transparent AI systems can help stakeholders understand the decision-making processes of algorithms, leading to increased trust and acceptance [240]. Furthermore, policies should consider the need for accountability and trustworthiness in AI deployment in agriculture [241]. Establishing guidelines for the level of supervision of AI tools and developing frameworks for handling issues related to the impact of AI tools on farmers and agricultural practices are essential. Moreover, regulations should aim to ensure that AI systems are explainable and interpretable, enabling stakeholders to comprehend the inner workings of these systems. At last policies and regulations are instrumental in promoting responsible AI deployment in agriculture by addressing data governance, privacy protection, and algorithm transparency. By developing comprehensive frameworks that encompass these aspects, policymakers can facilitate the ethical and sustainable integration of AI technologies into existing agricultural practices.

7. Future directions and challenges

The effective adoption of water-saving technologies is essential for the future of agriculture. Even while innovative approaches are showing promise, a number of obstacles prevent their widespread use. Significant challenges are presented by gaps in policy and socioeconomic limitations, such as insufficient financial resources and inadequate infrastructure. The use of these technologies is further impeded by farmers' lack of information and training. For there to be a resilient and sustainable agricultural future, these obstacles must be removed.

7.1. Remaining challenges and optimizing the impact of emerging technologies

Collaboration efforts are necessary to optimize the impact of emerging water-saving technologies and tackle the remaining challenges. Farmers need training, technical assistance, and extension services to appropriately develop their knowledge and skills in water-saving technologies [242]. Workshops, online courses, and certification programs can be provided for farmers to stay updated with technological advancements and acquire new skills. It is important to utilize investment funds, loans, and grants to support research and development in emerging technologies. Encourage venture capital investment in innovative startups and technologies are also crucial. Financial incentives, such as tax breaks, should be offered to companies further to support the implementation and integration of new technologies. This would be particularly beneficial for small and medium-sized businesses. Continually monitoring and assessing the impact of emerging technologies is essential for pinpointing areas for improvement and making necessary adjustments. This proactive approach enables timely interventions and ensures that technologies yield maximum positive effects. Sustainable frameworks and standards are essential for regulating and employing new technologies, minimizing potential dangers, and increasing trust between users and parties involved in innovative technology implementation. Collaboration between the public and private sectors and international cooperation will undoubtedly enhance the development and distribution of technological innovations, significantly expediting progress and ensuring substantial benefits.

7.2. Socio-economic implications and adoption barriers in different agricultural contexts

Several socioeconomic factors, including cultural norms, educational opportunities, and financial accessibility, impact the adoption of water-efficient technology. For example, in many rural smallholder areas, modern agricultural technology is limited by costs, inadequate infrastructure, and improper training. They find achieving maximum yield and resource efficiency hard without precision agriculture and remote sensing tools. Financial investment in futuristic agricultural technologies frequently does not have the required capital, which prevents small-scale farmers from being able to afford it [101,243]. Restricted loans and insurance are an added hindrance to adopting modern technologies and safeguarding themselves from potential risks. Nevertheless, the rural populations, limited knowledge, and literacy levels pose challenges to the uptake of these technologies, thereby significantly decreasing their positive effects. Cultural beliefs and social processes can affect agricultural systems where, at times, technological change is resisted. The resistance to change, based on tradition or the fear of disrupting social dynamics, is the main obstacle to innovation.

7.3. Policy regulations for adoption of efficient water use technologies

A well-defined policy framework is the key to promoting efficient water-use technology. Governments, in particular, pave the way for sustainable lifestyles by enacting laws, subsidies, and incentives. Through policies like water rate structures, irrigation water allocation systems, and conservation mandates, farming communities may be pushed to adopt water-saving technology. Additionally, the regulatory criteria for technology certification ensure that the technology is highly quality and reliable, instilling confidence in the stakeholders. Policymakers, scientists, and industry representatives should coordinate their efforts to develop policies that address agricultural conservation, production, and economic sustainability [244].

7.4. Emerging trends in water-saving technologies and research areas

Emerging trends in water-saving technologies and research areas will make it possible to achieve our desired future, as aquaponics and hydroponics are soil-less farming approaches that involve a recycled water system to grow plants in a controlled environment. Aquaponics systems save water and nutrients while increasing yield by integrating aquaculture and hydroponics [245]. In advanced leak detection systems, the water leakage wastes much water and is unnoticeable for extending the period. Modern leakage detection systems through sensors, acoustic technology, and data analytics can detect leaks in pipelines, infrastructure, and buildings at an early stage [246]. Thus, repairs can be done immediately with water conservation. Besides, desalination processes like reverse osmosis and distillation can eliminate salts and other pollutants from seawater or brackish water, converting the mass into quality water that is helpful in drinking or irrigation. The strides made in desalination technology are less energy-consuming and cheaper, further increasing its effectiveness in areas struggling with freshwater supply [247]. Rainwater harvesting systems are designed to capture and save the precipitating water for future use. Such systems can be as simple as a residential rain barrel to complex ones such as underground storage tanks and alternate water sources (like municipal water supply) for more extensive applications like commercial buildings and industrial companies. In greywater recycling systems; greywater, sinks, showers, and washing machines, wastewater can be purified for non-drinking purposes like irrigation, flushing toilets, and laundry. Recycling of greywater has reached adequate effectiveness and affordability, thus providing a sustainable alternative for water reuses [248].

8. Conclusion

The future of agriculture lies in adopting sustainable and resilient agricultural systems. Currently, smart farming processes are formalized, but a lack of knowledge about technological advancements could hinder the development of new automation capabilities in agriculture. Therefore, this paper presents a comprehensive study of state-of-the-art technologies to ensure efficient water use aimed at smart farming.

Recent literature has been reviewed, covering a variety of topics related to water usage in agriculture, including water pollution, irrigation, reuse, pipeline leaks, and livestock drinking water. The research community has investigated these areas, considering modern water management and monitoring techniques that rely on advanced technologies. Collaborative efforts are necessary to optimize the impact of emerging water-saving technologies and address remaining challenges. Continually monitoring and assessing the impact of these technologies is essential for identifying areas for improvement and making necessary adjustments. This proactive approach allows for timely interventions and ensures technologies yield maximum positive effects. Sustainable frameworks and standards are crucial for regulating and employing new technologies for sufficient water use, minimizing potential risks, and increasing trust among users and stakeholders involved in innovative technology implementation.

CRediT authorship contribution statement

Uttam Biswas Antu: Writing – original draft, Supervision, Project administration, Investigation, Conceptualization. **Md. Saiful Islam:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Conceptualization. **Sujat Ahmed:** Writing – review & editing, Visualization, Software, Resources, Investigation, Conceptualization. **Md. Arifuzzaman:** Writing – review & editing, Validation, Resources, Project administration, Investigation, Conceptualization. **Sawmitra Saha:** Writing – review & editing, Visualization, Validation, Software, Project administration, Investigation. **Puja Rani Mitu:** Writing – review & editing, Validation, Software, Project administration, Investigation. **Aditya Raj Sarkar:** Writing – review & editing, Visualization, Software, Methodology, Conceptualization. **Nor Aida Mahiddin:** Writing – review & editing, Visualization, Project administration, Formal analysis, Conceptualization. **Zulhilmi Ismail:** Writing – review & editing, Visualization, Supervision, Resources, Conceptualization. **Khalid A. Ibrahim:** Writing – review & editing, Validation, Software, Project administration, Conceptualization. **Abubakr M. Idris:** Writing – review & editing, Visualization, Resources, Project administration, Funding acquisition, Conceptualization.

Ethical approval

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Consent to publish

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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